Imperial College London

24 October 2024

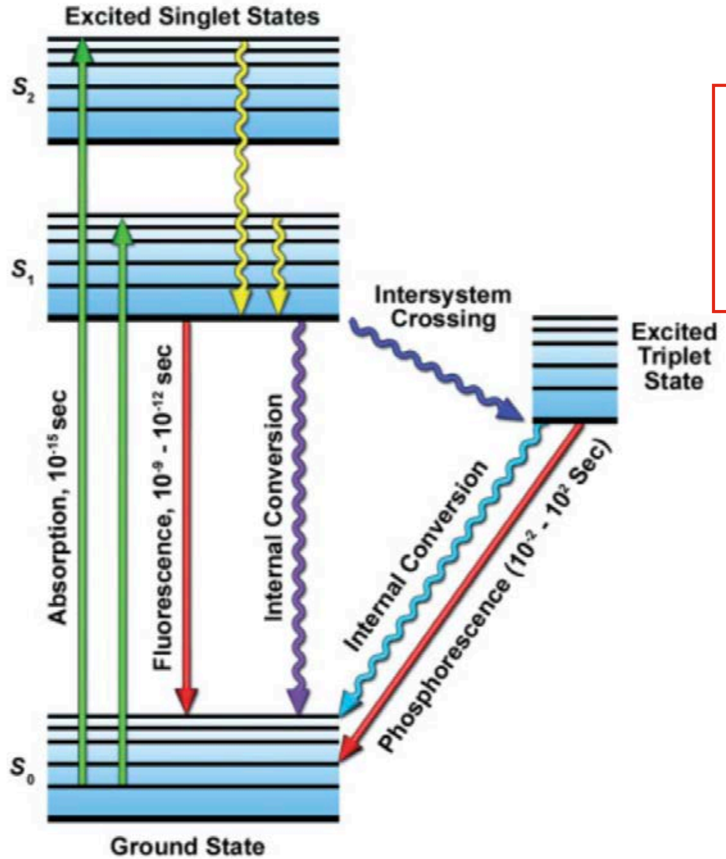
Advanced Imaging Technologies in Systems Biology and Biomedical Research

Periklis (Laki) Pantazis

What we covered

- What is fluorescence?
- How does a fluorescent microscope work?
- What are genetically encoded fluorescent proteins?

Physical basis of fluorescence: The Perrin-Jablonski diagram



chemically reactive:
- photobleaching
- production of damaging radicals

Physical basis of fluorescence: Important Parameters

molar extinction coefficient
potential of a fluorochrome to absorb photon quanta (given in optical density) at a reference wavelength (usually the absorption maximum)

quantum efficiency (QE)
fraction of absorbed photon quanta that is re-emitted by a fluorochrome as fluorescent photons

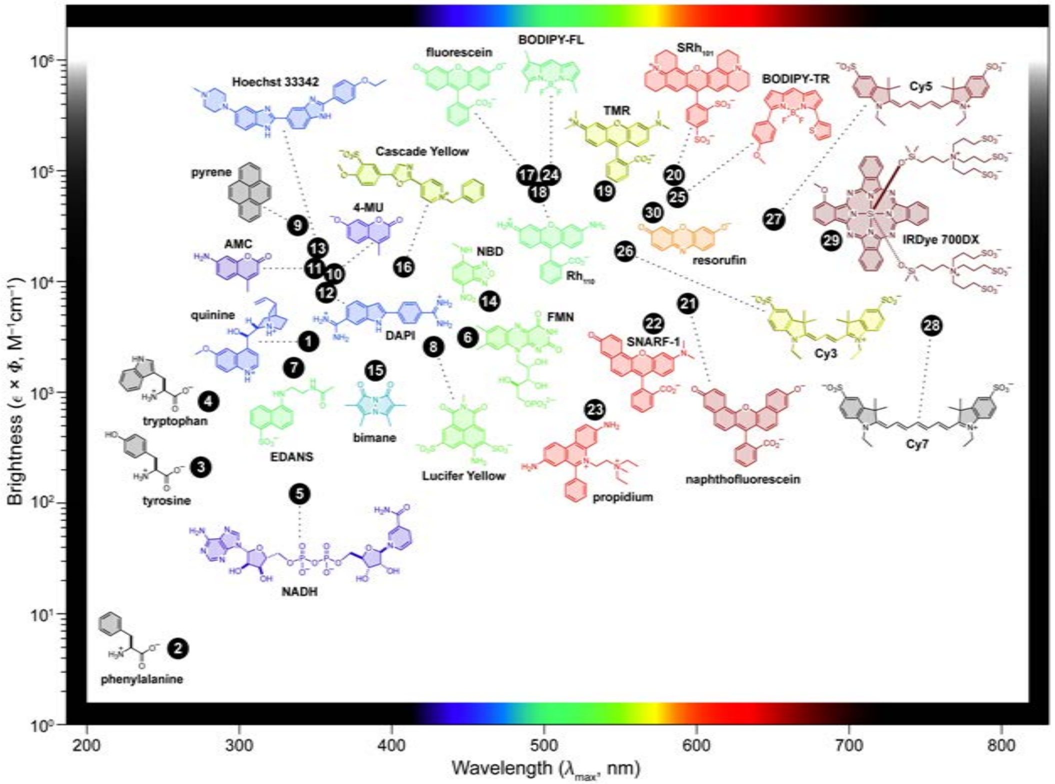
Quenching
reduction of quantum yield of a fluorochrome without changing its fluorescence emission spectrum; caused by interactions with other molecules including other fluorochromes

Photobleaching
permanent loss of fluorescence by a dye due to photon-induced chemical damage and covalent modification

Blinking
phenomenon that occurs during continuous excitation of a fluorescent molecule where emission transitions between “on” and “off” states

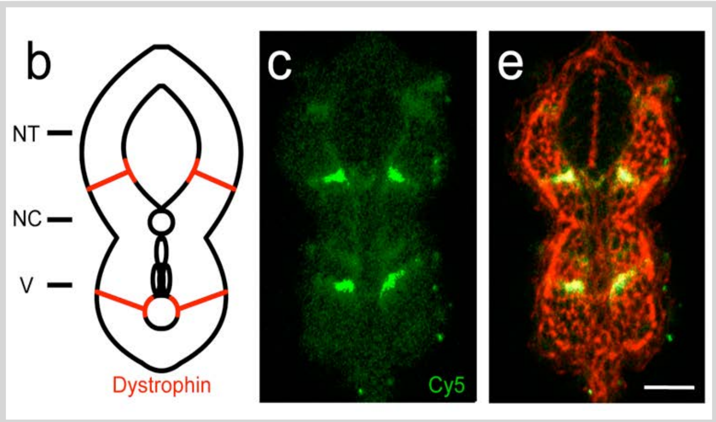
Properties of fluorescent dyes

BRIGHT - PHOTOSTABLE - COLOUR SELECTIVE - pH INSENSITIVE - WATER SOLUBLE



From Lavis, L.D., and Raines, R.T. 2008. ACS Chemical Biology 3:142–155

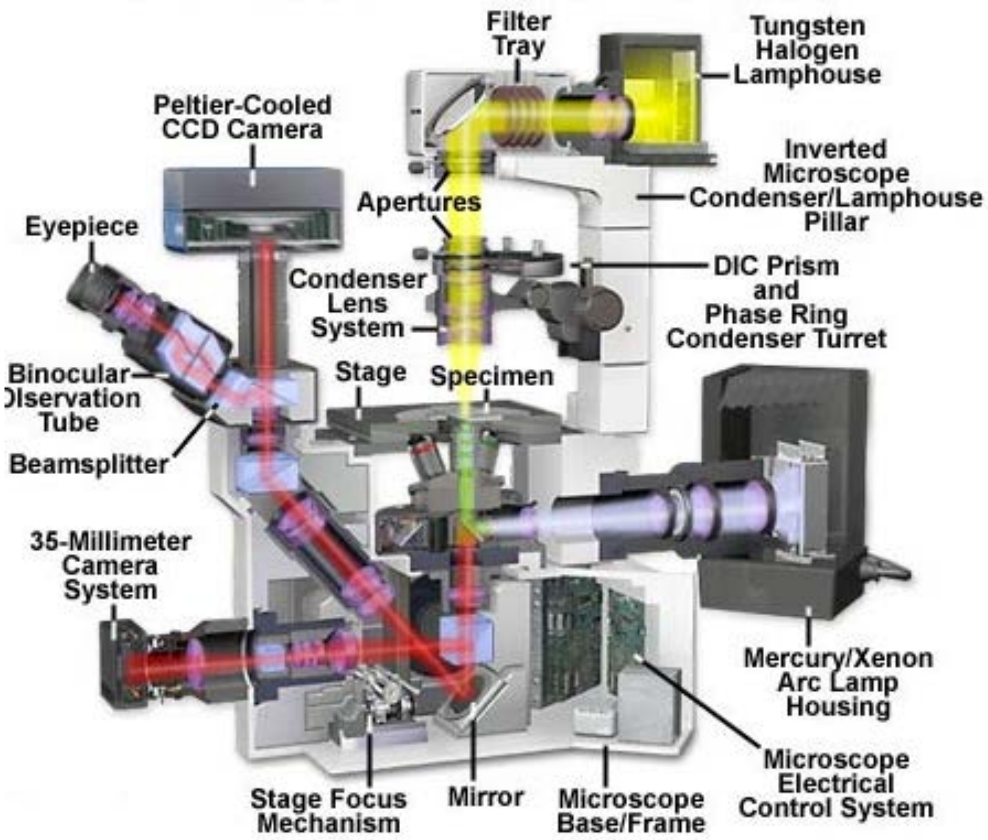
Autofluorescence of endogenous molecules



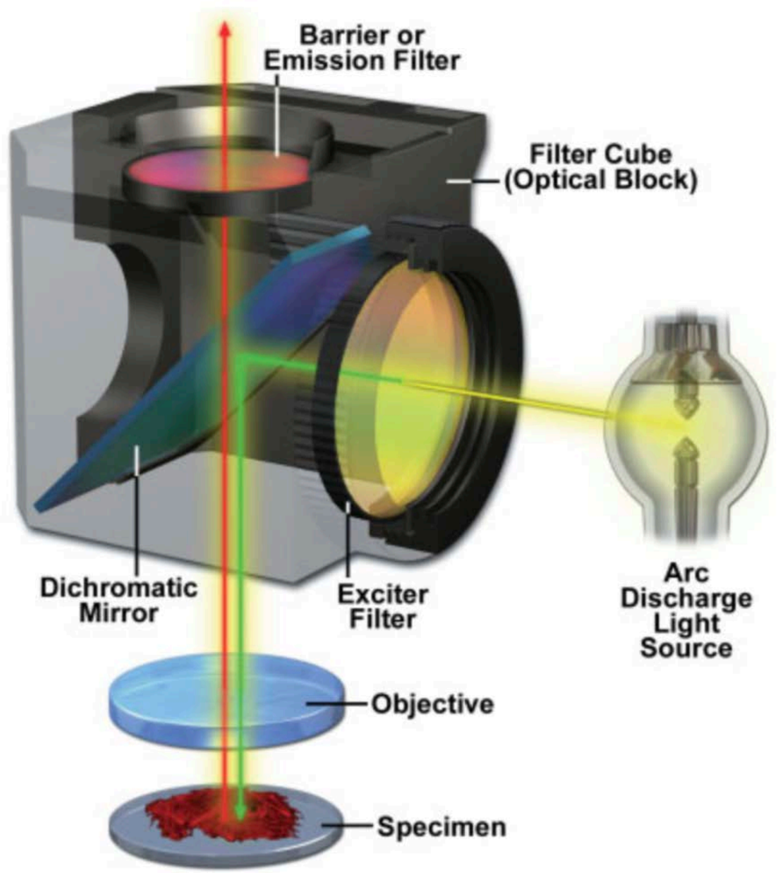
Some common sources of autofluorescence
B vitamins
flavin nucleotides (FAD and FMN)
reduced pyridine nucleotides (NADH and NADPH)
fatty acids
porphyrins
serotonin
catecholamines

Proc Natl Acad Sci U S A. (2010); 107(33):14535-40.

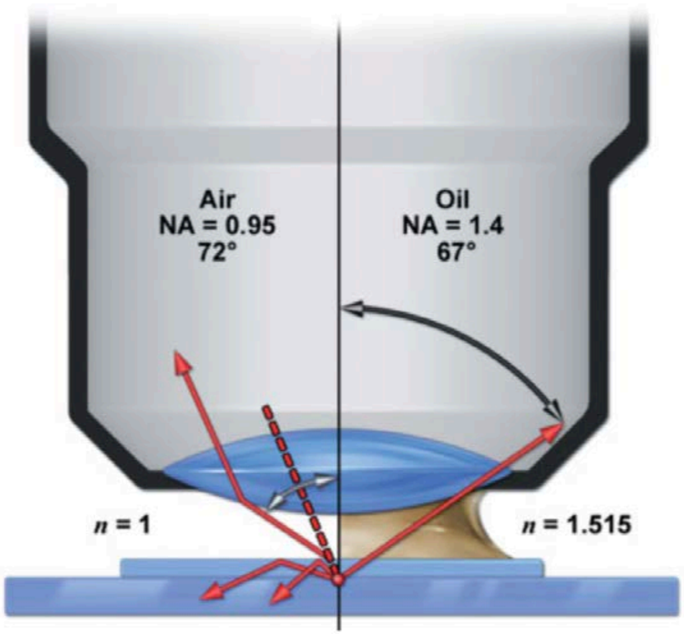
Arrangement of a fluorescent microscope



Arrangement of filters in a filter cube



Spatial resolution in fluorescent microscopy



for non-luminous points that are examined by bright-field microscopy in transmitted light where the condenser NA is ≥ the objective NA, the resolving power of the microscope is defined as

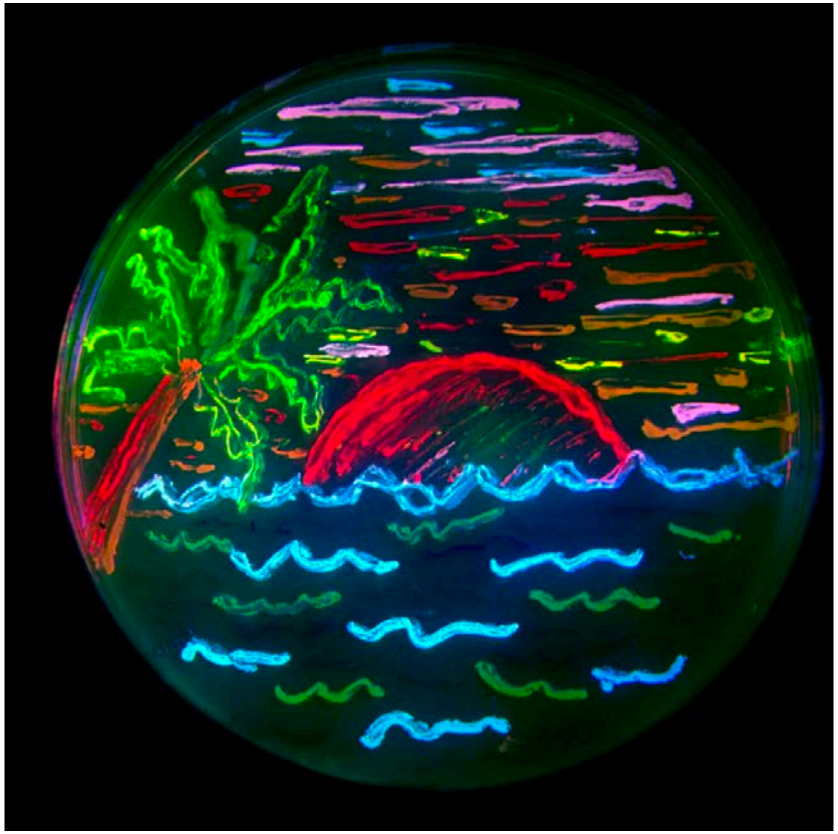
$$d = 0.61\ \lambda / NA$$

In the light microscope, the angular aperture is described in terms of the numerical aperture (NA), as:

$$NA = n \sin(\theta)$$

d = minimum resolved distance in μm
λ = wavelength in μm
NA = numerical aperture of the objective lens
n = refractive index of the medium between the lens and the object

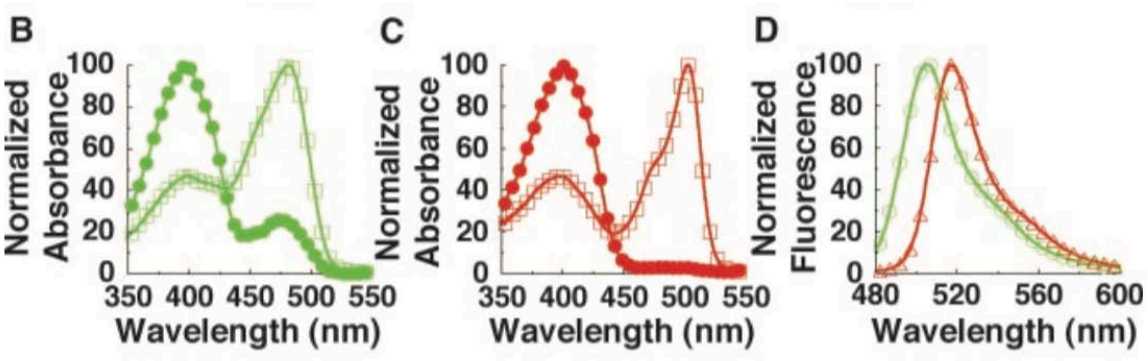
Differently coloured fluorescent proteins

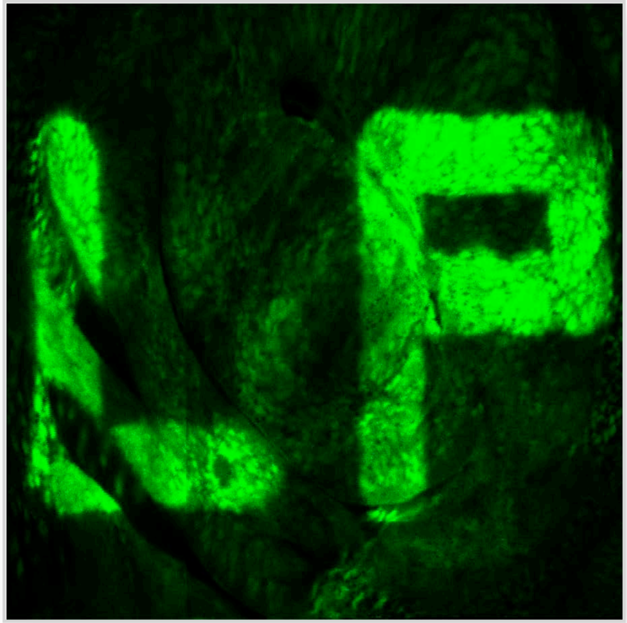


Protein[a]	Color[b]	Excitation (nm)	Emission (nm)	Brightness[c]	Photostability[d]	Filter Set[e]
EBFP2	Blue	383	448	18	++	DAPI
mCerulean	Cyan	433	475	17	++	CFP
mTurquoise	Cyan	433	474	25	+++	CFP
mTFP1	Teal	462	492	54	+++	CFP
mEGFP	Green	488	507	34	++++	FITC/GFP
mEmerald	Green	487	509	39	++++	FITC/GFP
mVenus	Yellow	515	528	53	++	FITC/YFP
mCitrine	Yellow	516	529	59	++	FITC/YFP
mKO2	Orange	551	565	40	+++	TRITC
tdTomato	Orange	554	581	95	+++	TRITC
TagRFP	Orange	555	584	48	++	TRITC
mApple	Orange	568	592	37	+++	TRITC
mCherry	Red	587	610	17	+++	TxRed
mKate2	Far-Red	588	633	25	++	TxRed
mPlum	Far-Red	590	649	3.2	+++	TxRed
mNeptune	Far-Red	600	650	13	++++	Cy5

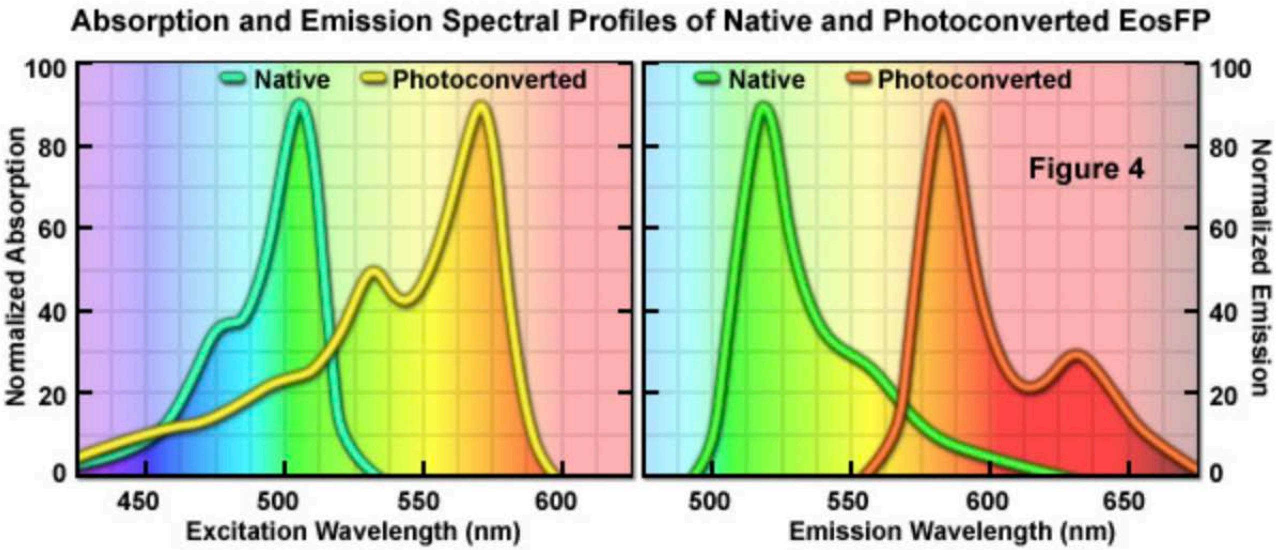
[a] Common literature abbreviation.
[b] Spectral class.
[c] Product of the molar extinction coefficient and the quantum yield $(mM \times cm)^{-1}$.
[d] Relative to mEGFP (++++).
[e] Recommended filter set.

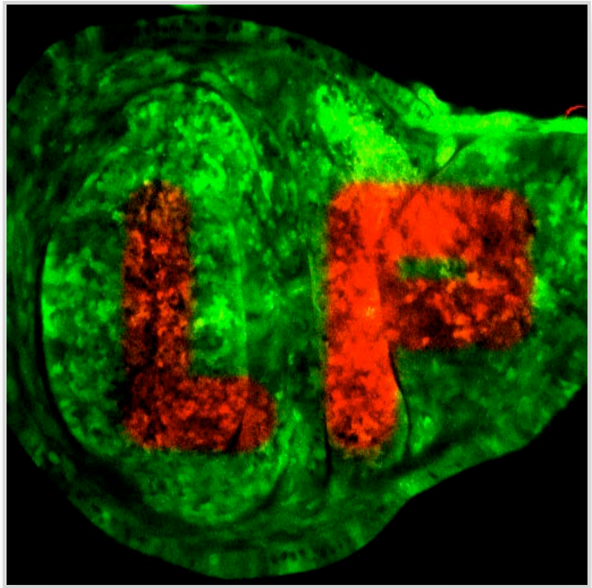
Photoactivatable fluorescent proteins





Photoconvertible fluorescent proteins





What we will cover

- How to visualise fluorescent probes in thick specimen?
- How to achieve deep tissue imaging?

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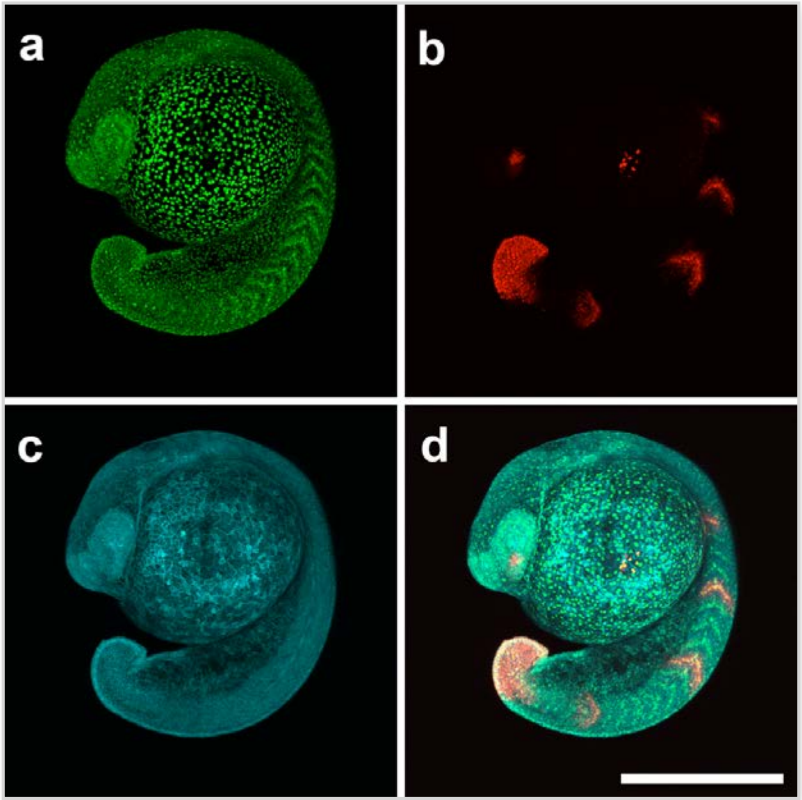
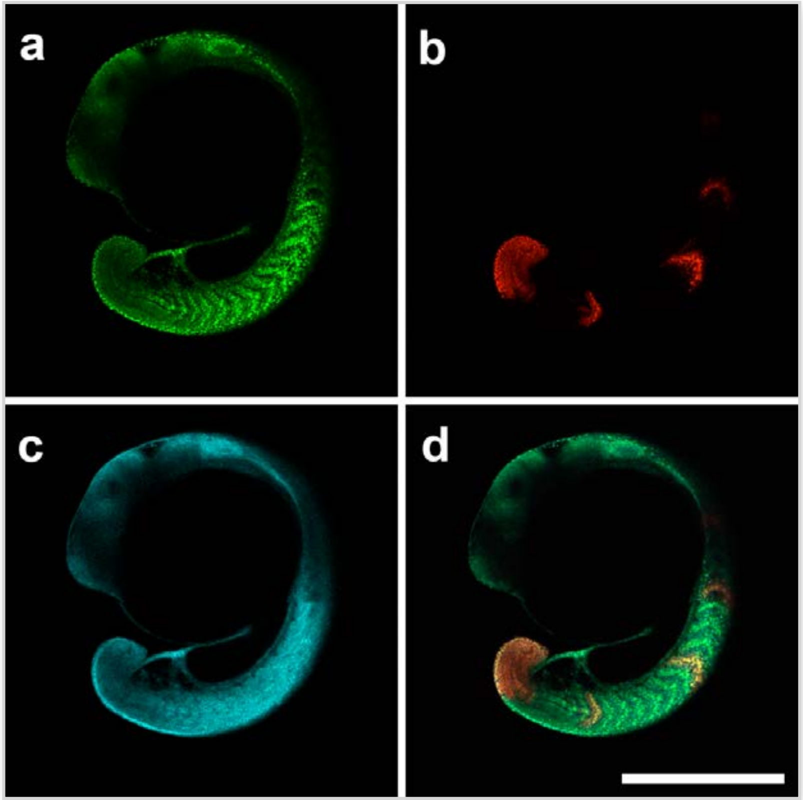
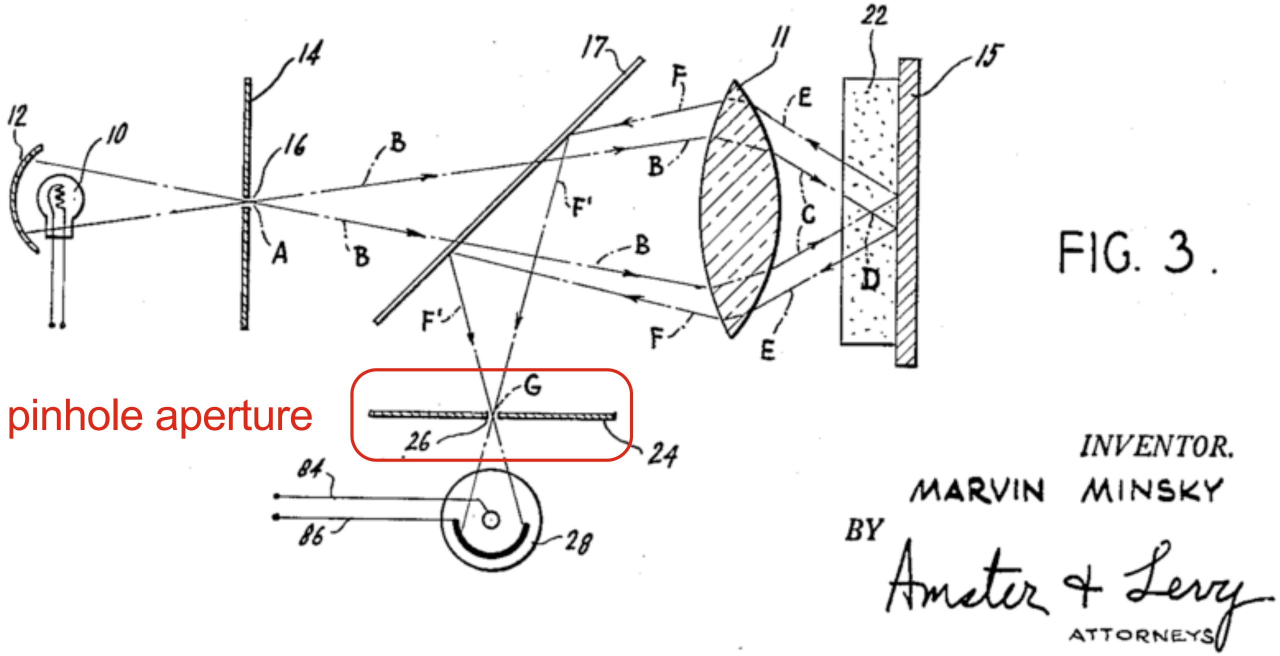
- How to achieve deep tissue imaging

Optical sections in fluorescent microscopy





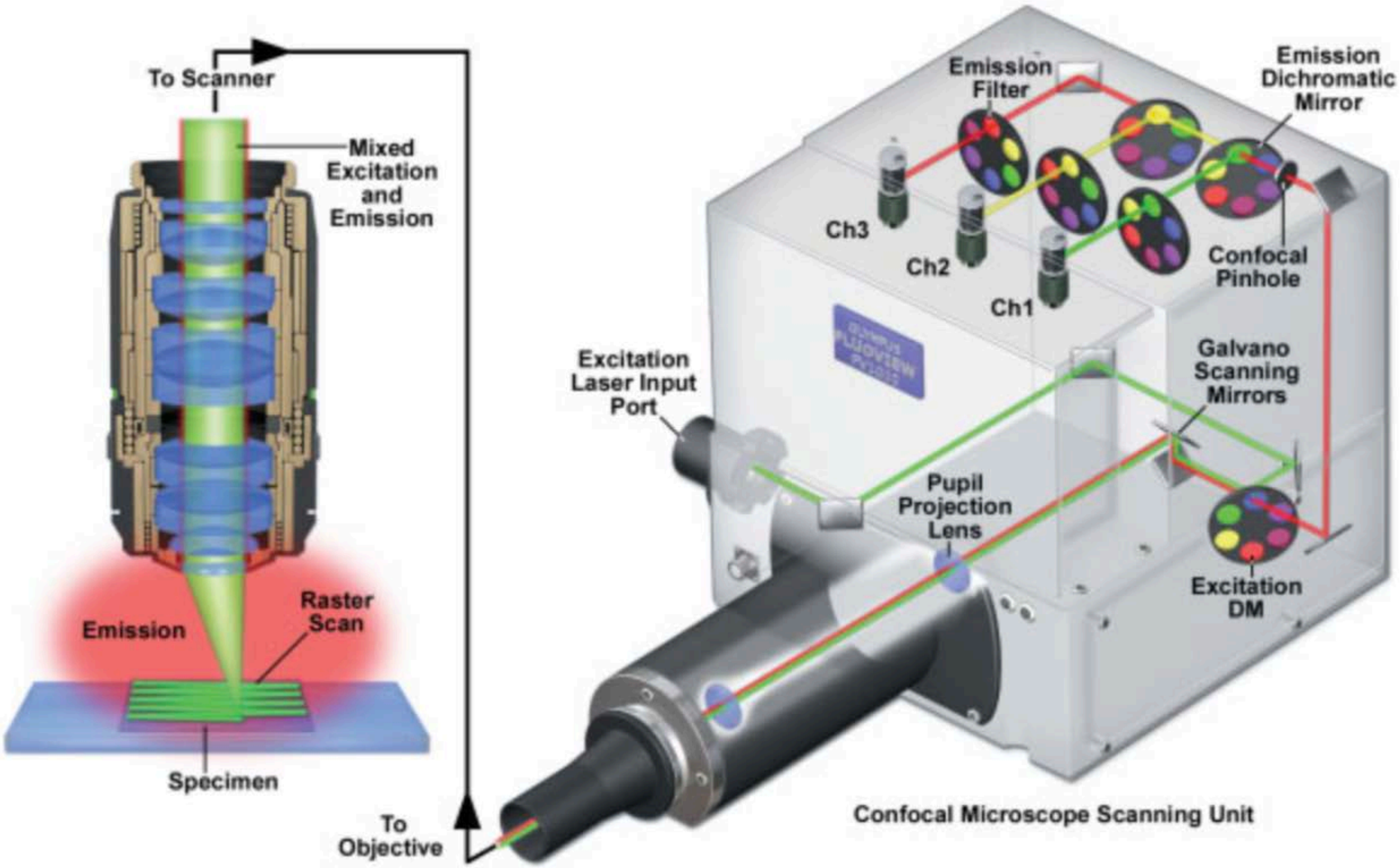
Confocal optical section of a specimen - the principle



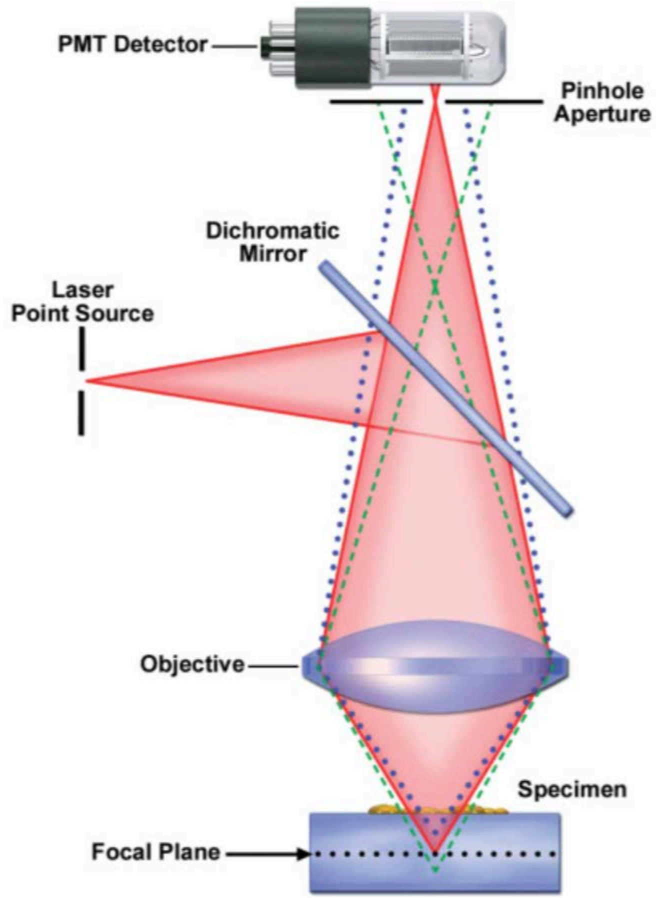
Pinhole Aperture: accepts fluorescent photons from the illuminated focused spot, but largely excludes fluorescence signals from objects above and below the focal plane

Basic components of a confocal laser scanning microscope (CLSM)

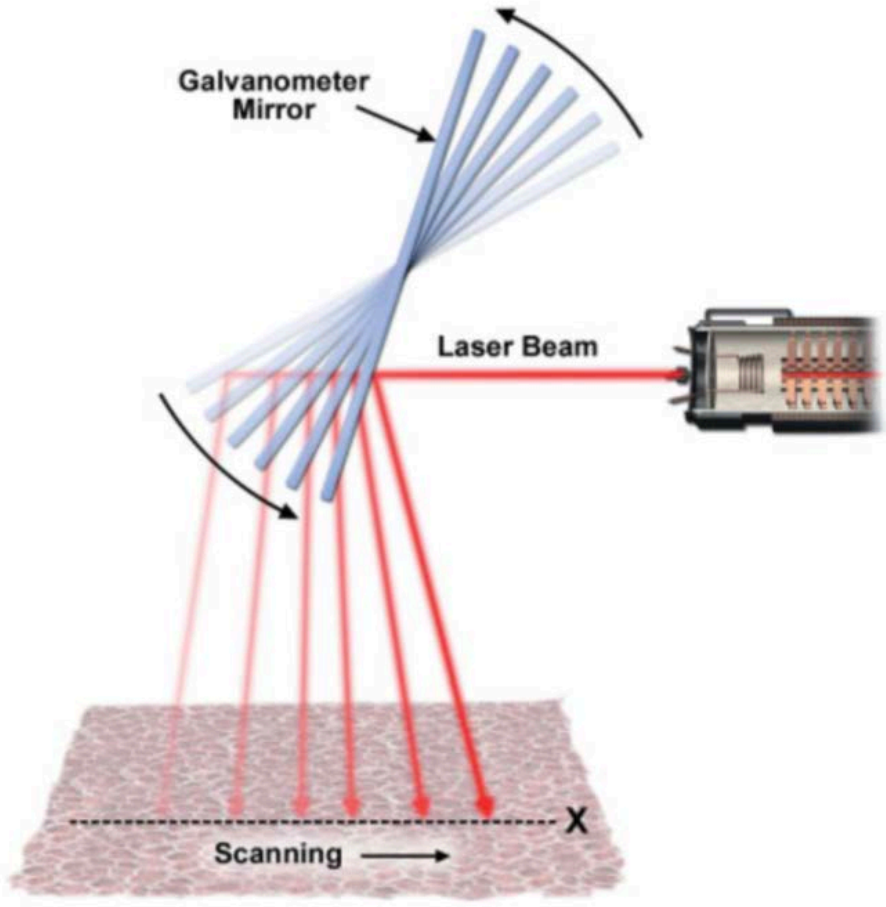
Optical pathway in a confocal microscope point-scanning unit



The confocal principle in epi-fluorescence laser scanning microscopy



The scan-control mechanism in a CLSM



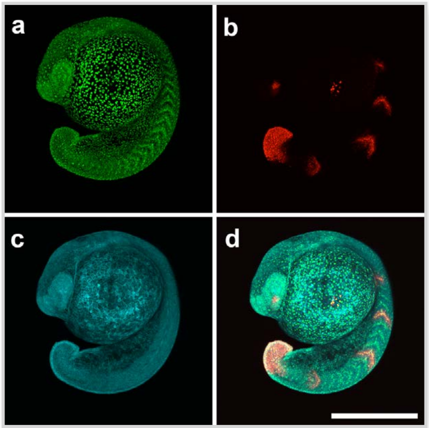
Lasers for fluorophore excitation in CLSM

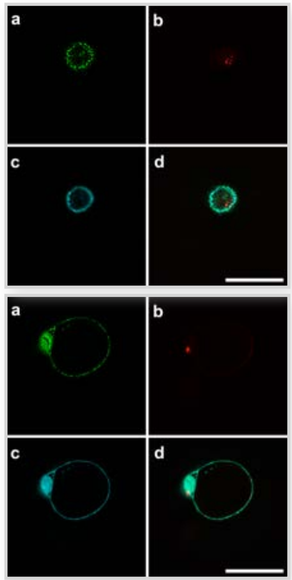
	Wavelength (nm)			
Laser Type	Violet	Blue	Green	Red
Argon-ion		458, 477, 488	515	
Helium-neon			543	594, 633
Diode	405	440, 473, 488	539, 559	638, 647
Diode-pumped solid state			532, 561	640

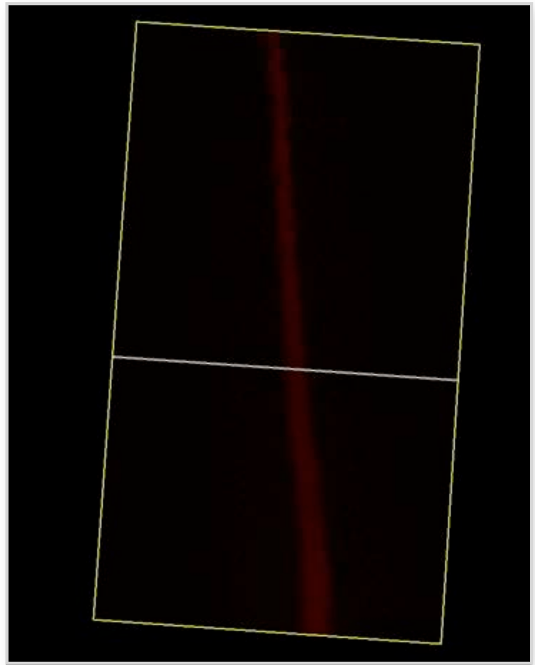
Wide-field fluorescence system versus CLSM

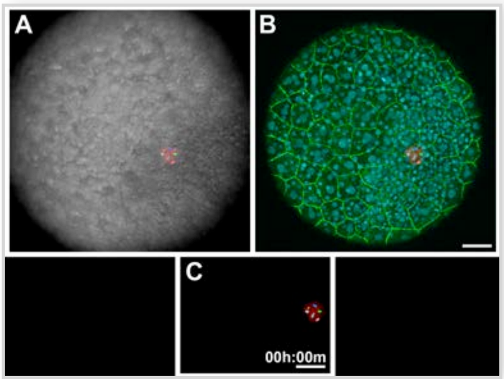
Fixed Tissue	Which microscope method would you use to evaluate whether proteins co-localise?
Live Tissue	Which method would you use for tracking fluorescent cells?

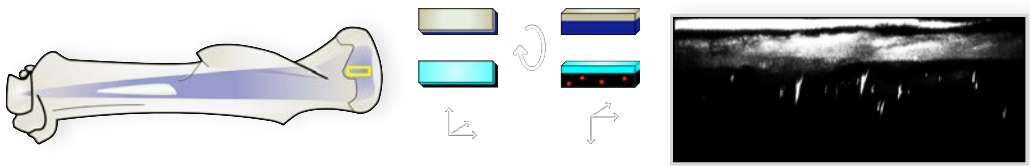
Advantages of confocal over wide-field fluorescence microscopy











Spatial resolution in CLSM

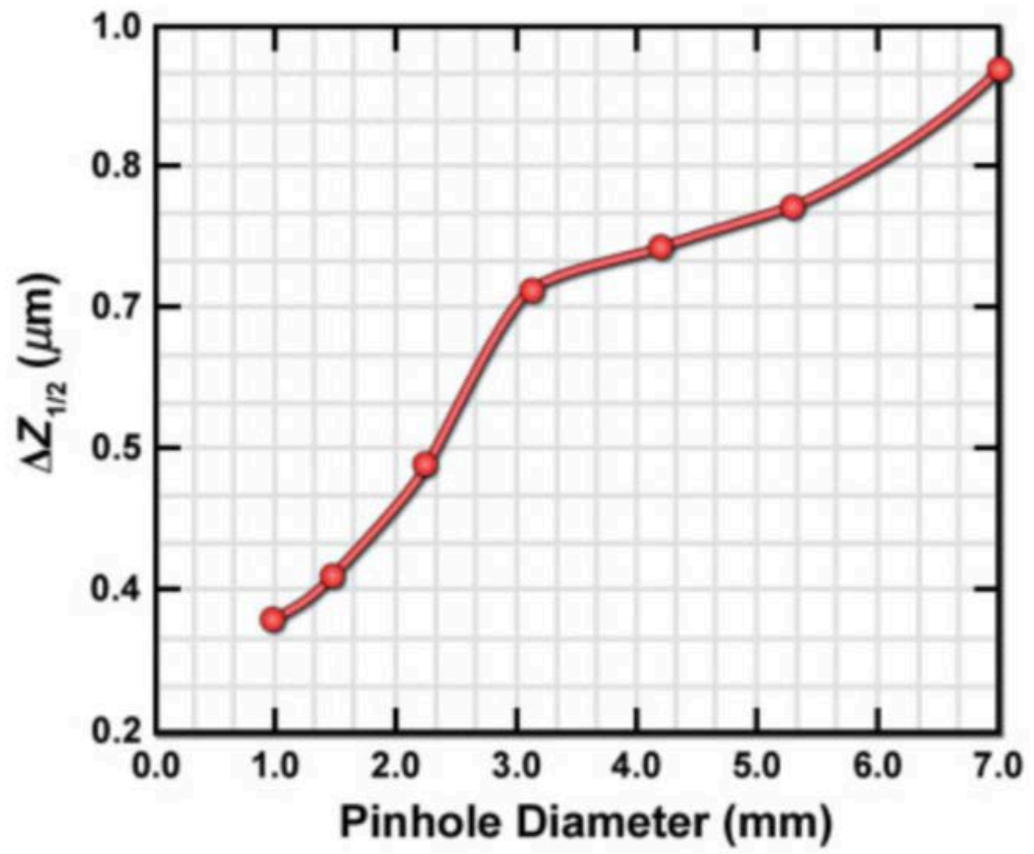
the smallest distance that can be resolved using confocal optics is proportional to $(1/\lambda 1 + 1/\lambda 2)$, and the parameters of wavelength and numerical aperture figure twice into the calculation for spatial resolution

$$d_{x,y} = 0.4\, \lambda / NA$$

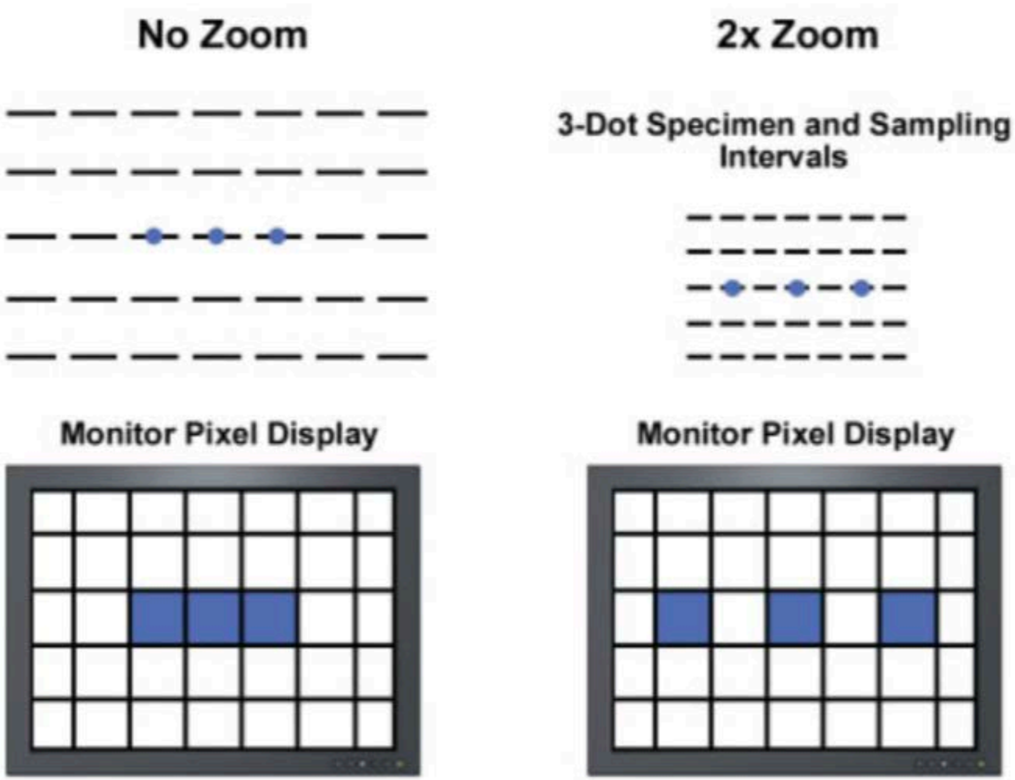
$$d_z = 1.4n\, \lambda / NA2$$

d = minimum resolved distance in μm
λ = wavelength in μm
NA = numerical aperture of the objective lens
n = refractive index of the medium between the lens and the object

Confocal microscope pinhole diameter versus optical section thickness



Effect of electronic zoom on electronic sampling and image resolution



Electronic zoom is used for two reasons:

1. to reduce pixel size and maintain spatial resolution, and
2. to magnify the image for display and printing.

Increasing the zoom factor reduces the area scanned on the specimen and slows the rate of scanning.

The increased number of sampling intervals (pixels) along a comparable length in the specimen increases the spatial resolution and magnifies the display on the monitor.

Electronic adjustments and considerations for confocal fluorescence imaging: Your selection does matter!!!

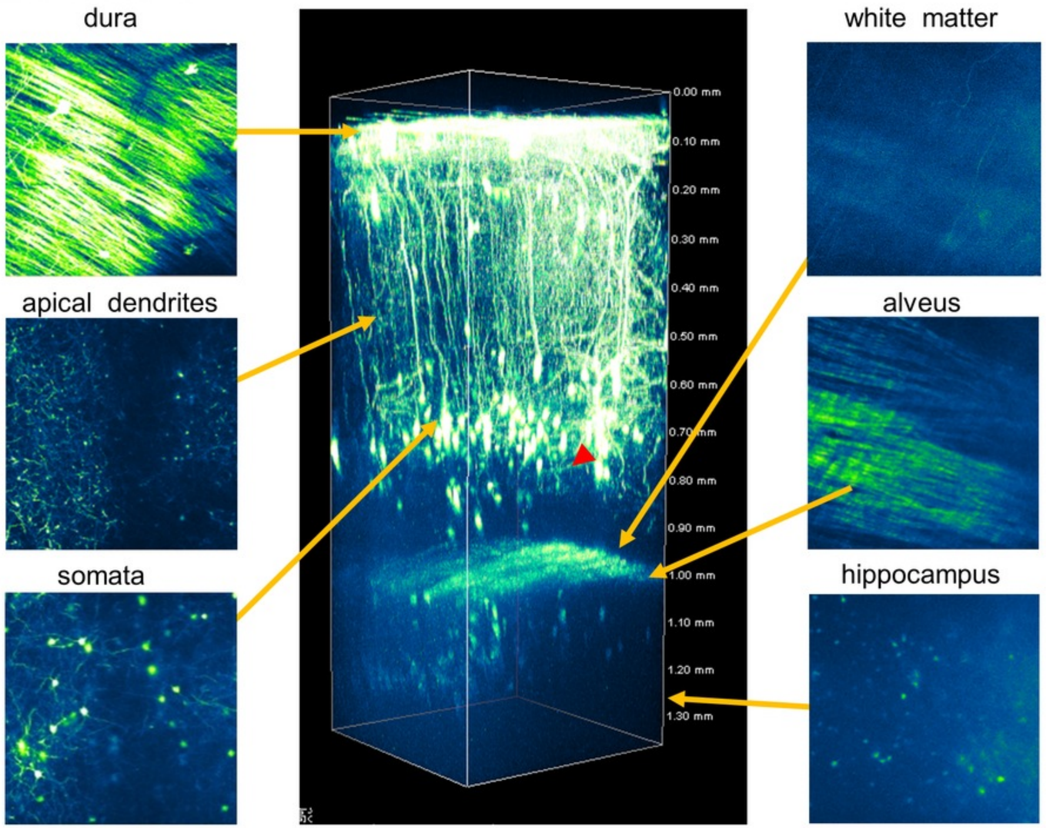
Parameter	Effect
Pinhole diameter	Intensity Spatial resolution
Zoom	Intensity Spatial resolution
Scan rate	Intensity Temporal resolution
Objective NA	Intensity Spatial resolution

What we will cover

- How to visualise fluorescent probes in thick specimen?

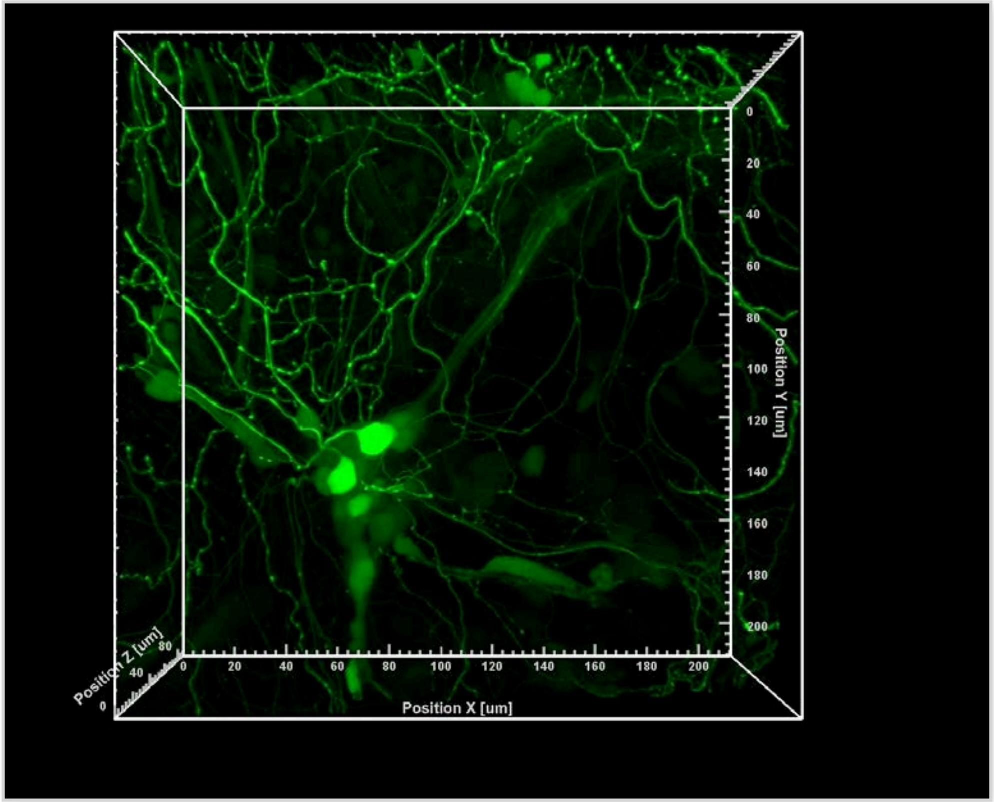
- **How to achieve deep tissue imaging**

Bright, high contrast, fluorescence images even at great depths in thick specimens



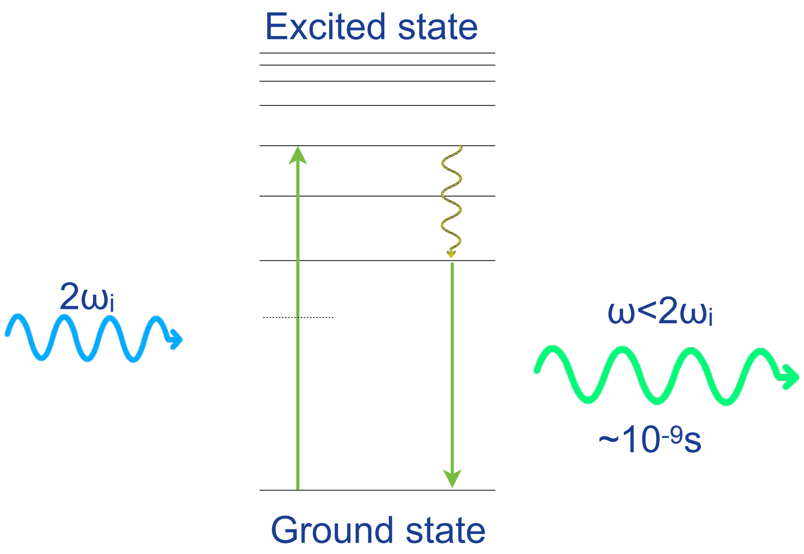
Brain structures of hundreds of microns in depth!

How to unravel complex structural information of individual neurons?

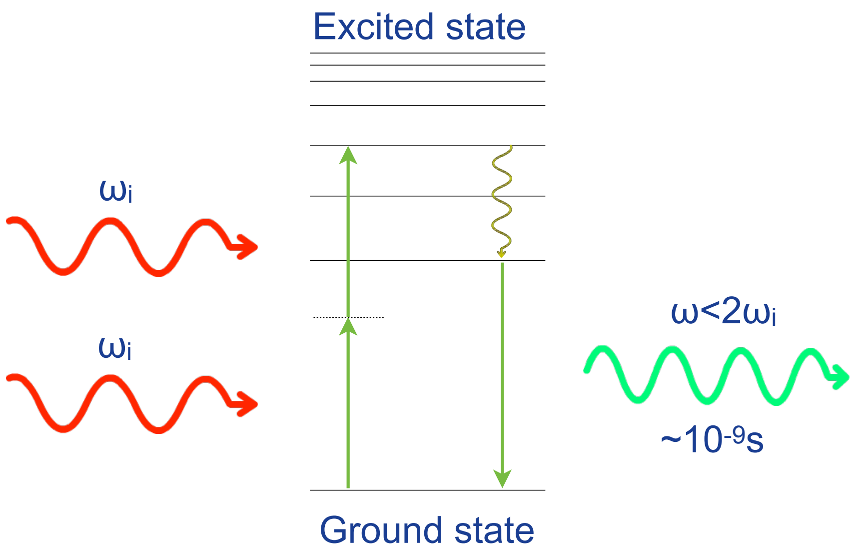


Surrounded by tightly packed neighbouring cells!

1-Photon excited fluorescence versus 2-Photon excited fluorescence

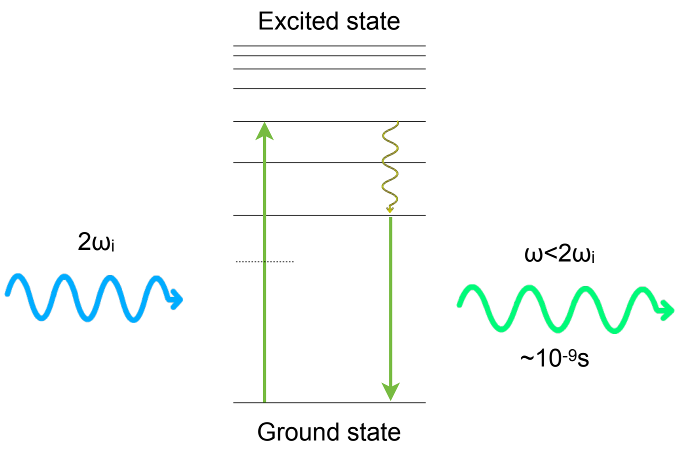


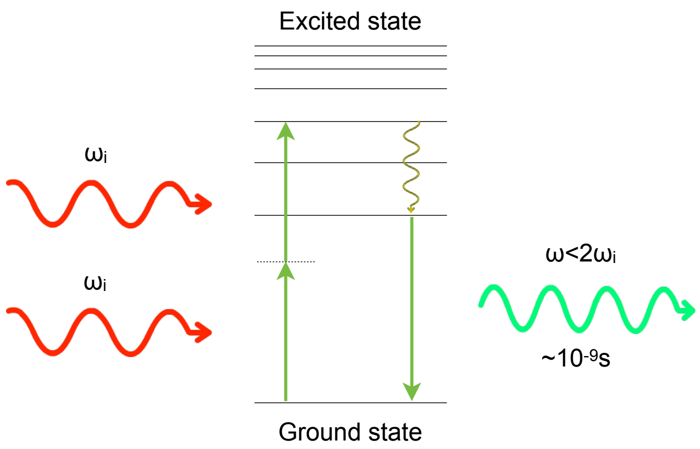
1-Photon excited fluorescence



2-Photon excited fluorescence

1-Photon excited fluorescence versus 2-Photon excited fluorescence





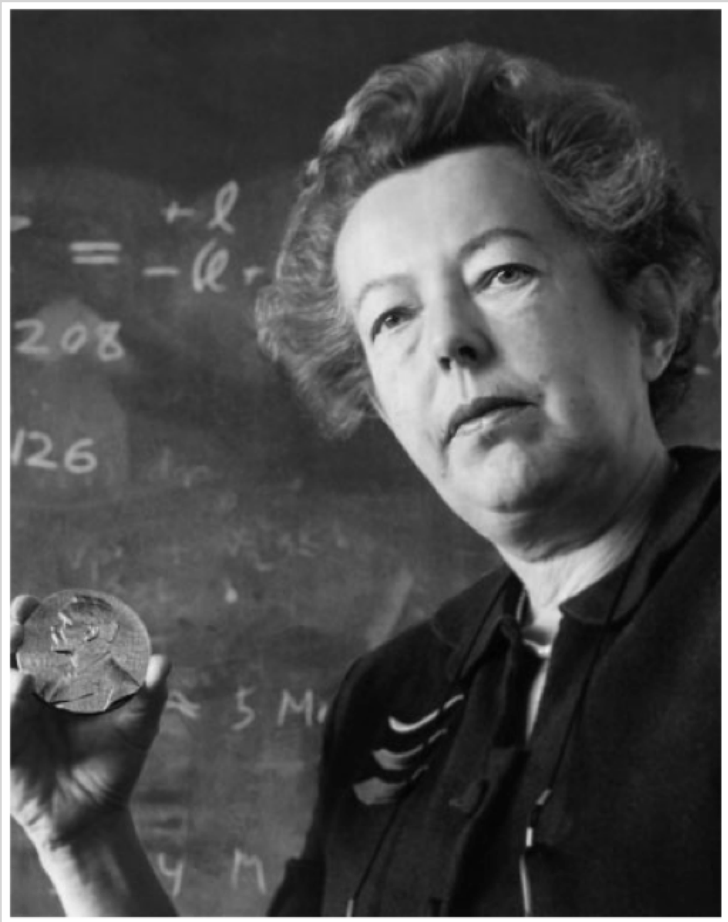
1-Photon excited fluorescence

- Limited penetration
- Increased phototoxicity above and below
- Excitation not restricted to the focal point
- pinhole required
- high probability - medium intensity in the focal plane

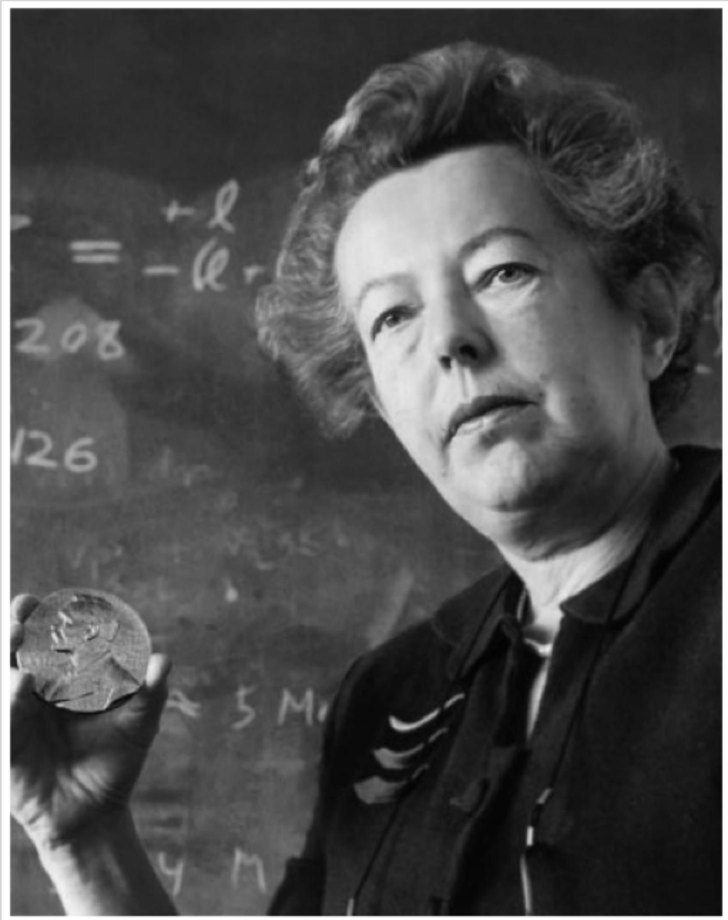
2-Photon excited fluorescence

- Deeper penetration
- Reduced phototoxicity above and below
- Excitation restricted to the focal point
- pinhole not required
- low probability - high intensity in the focal plane -

Maria Göppert-Mayer (1906–1972)



Maria Göppert-Mayer (1906–1972)



<u>Probability of molecular excitation</u>

a) molecular two-photon cross-section, σ

σ = areas of two excitation photons

x

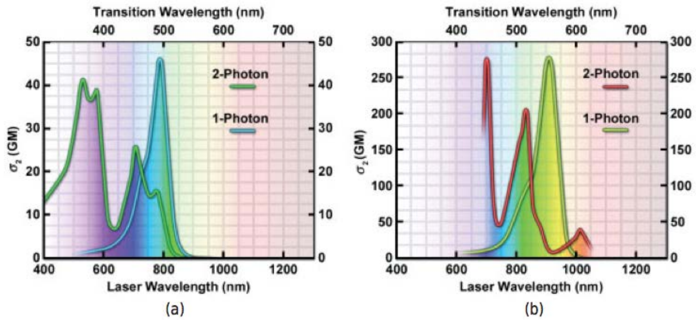
the time within which a target molecule simultaneously absorbs two photons and enters an excited state ($\sim 10^{-18}$ s)

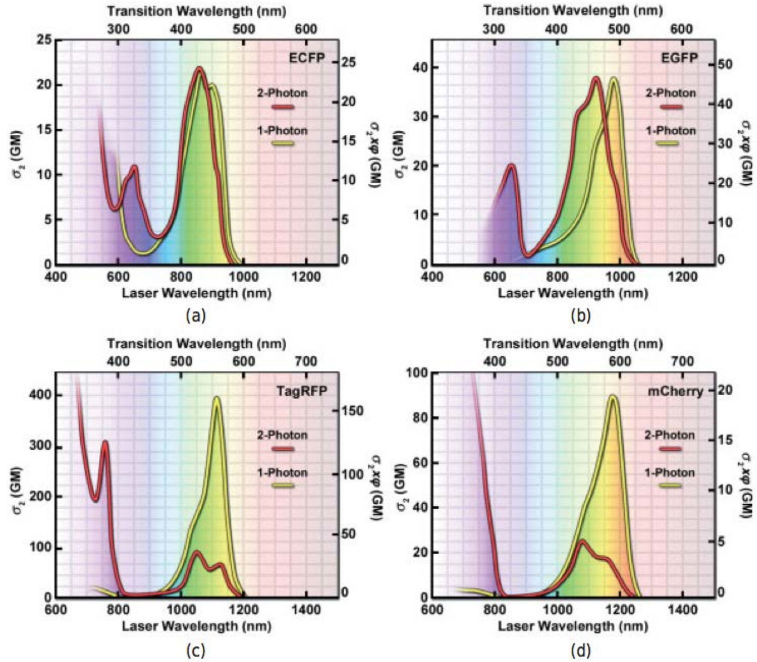
b) two-photon *action cross-sections*

the product of σ and QE, which are given for a particular wavelength in *GM units*

$1\ \mathrm{GM} = 10^{-50}\,\mathrm{cm}^4\cdot\mathrm{s}\cdot\mathrm{photon}^{-1}$

Two-photon excitation spectra for common probes





To roughly estimate the wavelength required to stimulate a fluorescent dye by the two-photon process, one just doubles the one-photon absorption maximum

Bright, high contrast, fluorescence images even at great depths in thick specimens: two-photon microscopy

W. Webb

W. Denk

Two-Photon Laser Scanning Fluorescence Microscopy

WINFRIED DENK,* JAMES H. STRICKLER, WATT W. WEBB

Molecular excitation by the simultaneous absorption of two photons provides intrinsic three-dimensional resolution in laser scanning fluorescence microscopy. The excitation of fluorophores having single-photon absorption in the ultraviolet with a stream of strongly focused subpicosecond pulses of red laser light has made possible fluorescence images of living cells and other microscopic objects. The fluorescence emission increased quadratically with the excitation intensity so that fluorescence and photobleaching were confined to the vicinity of the focal plane as expected for cooperative two-photon excitation. This technique also provides unprecedented capabilities for three-dimensional, spatially resolved photochemistry, particularly photolytic release of caged effector molecules.

WE REPORT THE APPLICATION OF two-photon excitation of fluorescence (*1*) to laser scanning microscopy (LSM) (*2*). In LSM, the laser is focused to a diffraction-limited beam waist <1 μm in diameter and is raster-scanned across a specimen. Confocal LSM provides depth discrimination and improves spatial resolution within the plane of focus by forming the image through the same scanning optics used for illumination and through a pinhole placed in front of the detector, which acts as a spatial filter to select emission from the plane of focus (*2*). Very good laser scanning micrographs have been obtained with the use of fluorescent markers that absorb and emit visible light (*3*). Confocal scanning images with fluorophores and fluorescent chemical indicators that are excited by the ultraviolet (UV) part of the spectrum have not appeared, however, largely because of the lack of suitable microscope lenses, which must be chromatically corrected and transparent for both absorption and emission wavelengths (*4, 5*). Anticipated photodamage to living cells and the limitations of UV lasers have also inhibited this approach.

Two-photon molecular excitation (*1, 6*) is made possible by the very high local instantaneous intensity provided by the tight focusing in an LSM combined with the temporal concentration of a femtosecond pulsed laser. With a colliding-pulse, mode-locked dye laser (*7*) (CPM) producing a stream of pulses with a pulse duration (τ_p) of about 100 fs at a repetition rate (f_p) of about 80 MHz, the probability becomes appreciable for a dye molecule to absorb two long-wavelength (λ_{CPM}) photons simultaneously, thus combining their energy in order to reach its excited state. Assuming a typical two-photon cross section (*8*) of $\delta = 10^{-58}$ m^4-s per photon with the above pulse parameters and the beam focused by a lens of numerical aperture $A \approx 1.4$, we calculated that an average incident laser power (p_0) of ~50 mW would saturate the fluorescence output at the limit of one absorbed photon pair per pulse per fluorophore (*9*). The fluorescence emission could be increased, however, if the pulse repetition frequency were increased to the inverse fluorescence

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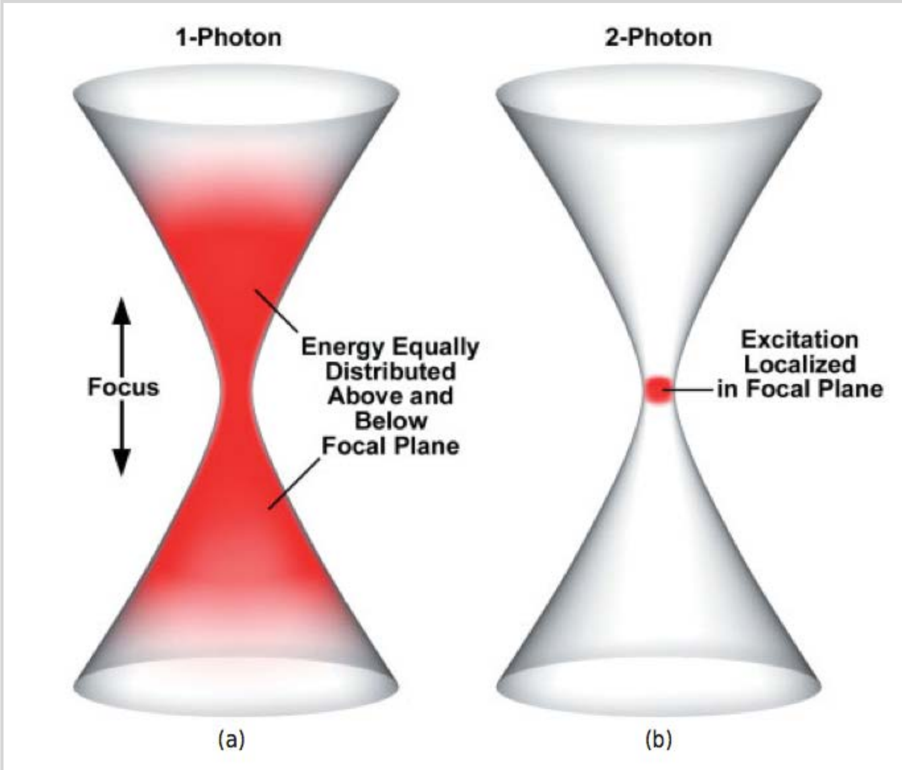
Two-Photon Microscopy:
near-infrared (ir) laser & localised excitation of fluorescence

Localisation of excitation by one-photon excitation versus two-photon excitation

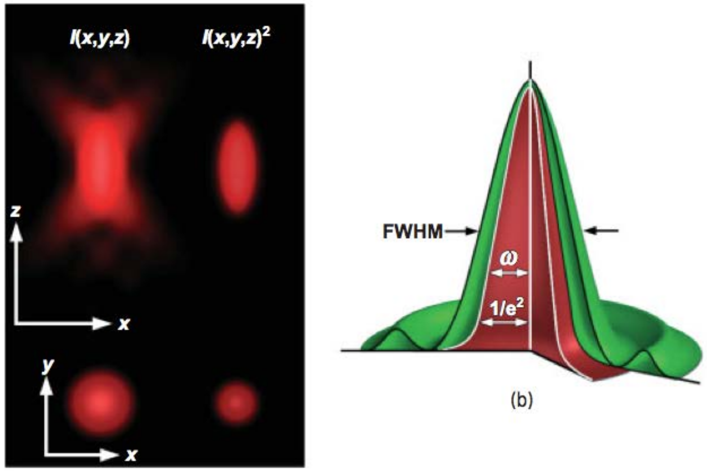


Factors governing total fluorescence present at different distances from the focal plane by one-photon and two-photon mechanisms

Terms	1 Photon	2 Photon
Absorption rate ϕ	$\phi \sim I$	$\phi \sim I^2$
Intensity I	$I \sim 1/z^2$	$I \sim 1/z^2$
Rewriting ϕ	$\phi \sim I/z^2$	$\phi \sim I/z^4$
Area A of illuminated spot at each distance z from the focal plane[a]	$A \sim z^2$	$A \sim z^2$
Total amount of fluorescence Φ at each focal plane	$\Phi \sim \phi A \sim 1/z^2 \bullet z^2 = 1$ Φ is constant in each cross-sectional area at every distance z from the focal plane	$\Phi \sim \phi A \sim 1/z^4 \bullet z^2 = 1/z^2$ Φ decreases as the square of the z distance, which localizes fluorescence to the focal plane

A key difference between one-photon and two-photon excitation is that the absorption rate (φ) is directly proportional to intensity I in one-photon excitation, but is proportional to I^2 in two-photon excitation and is thus a nonlinear process.

Two-photon imaging resolution is proportional to the square of the intensity point spread function (IPSF2)



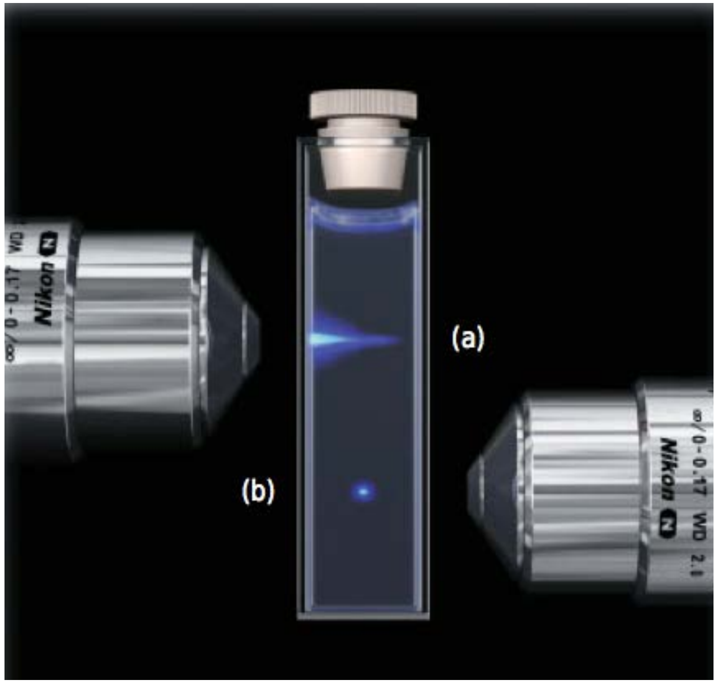
(a) (b)

For objective NA > 0.7, the lateral and axial resolutions can be calculated as:

$\omega_{xy} = 0.325\lambda/\sqrt{2}\ NA0.91$
$\omega_{z} = [0.532\lambda/\sqrt{2}]\cdot[1/(n-\sqrt{(n2 - NA2)}]$

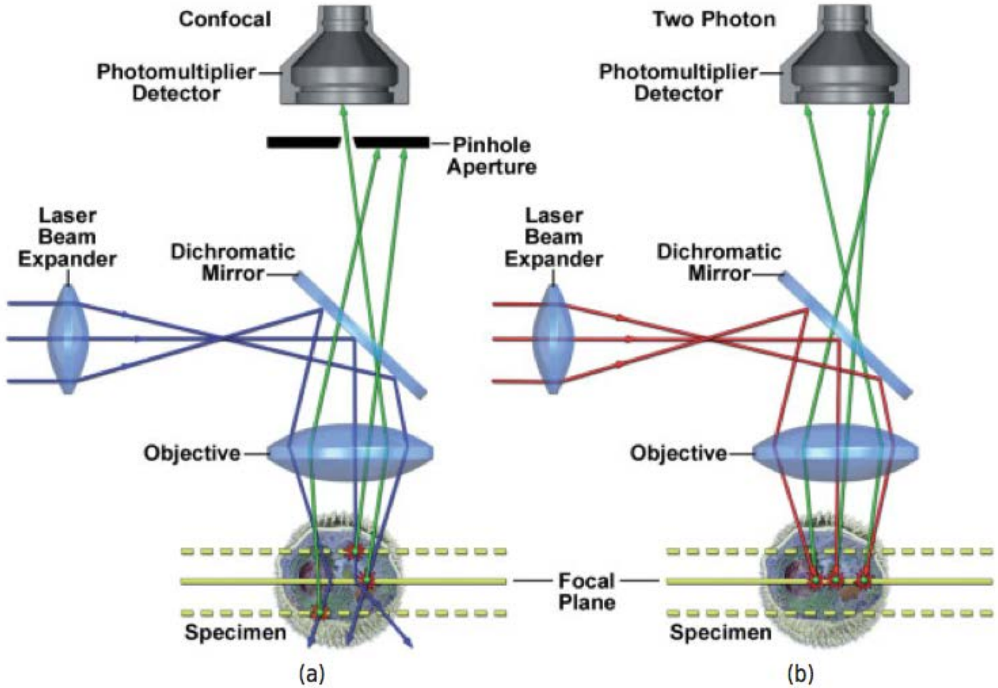
ω = 1/e times the value of the respective lateral and axial radii of the two-photon point spread function (IPSF2)
n = the refractive index

Localisation of excitation by one-photon excitation versus two-photon excitation



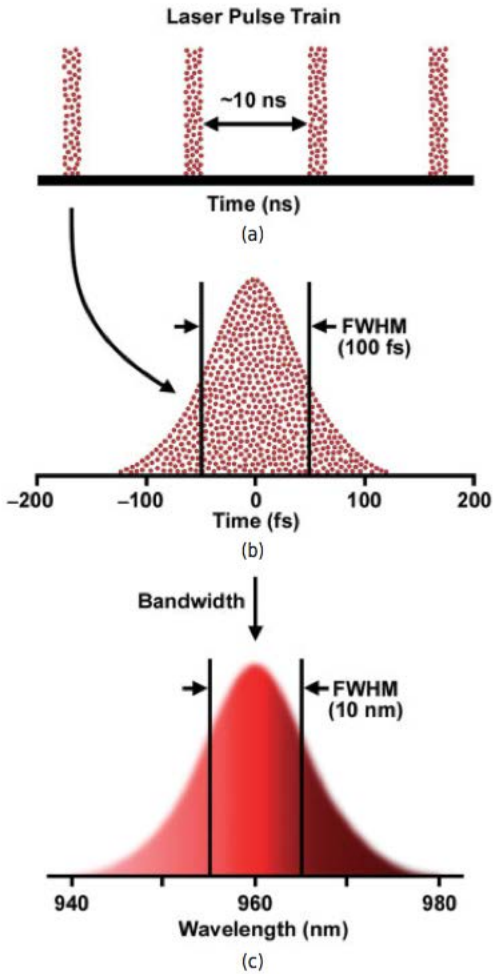
Two-photon excitation diminishes with the 4th power of the distance from the focal plane

Confocal imaging versus two-photon imaging

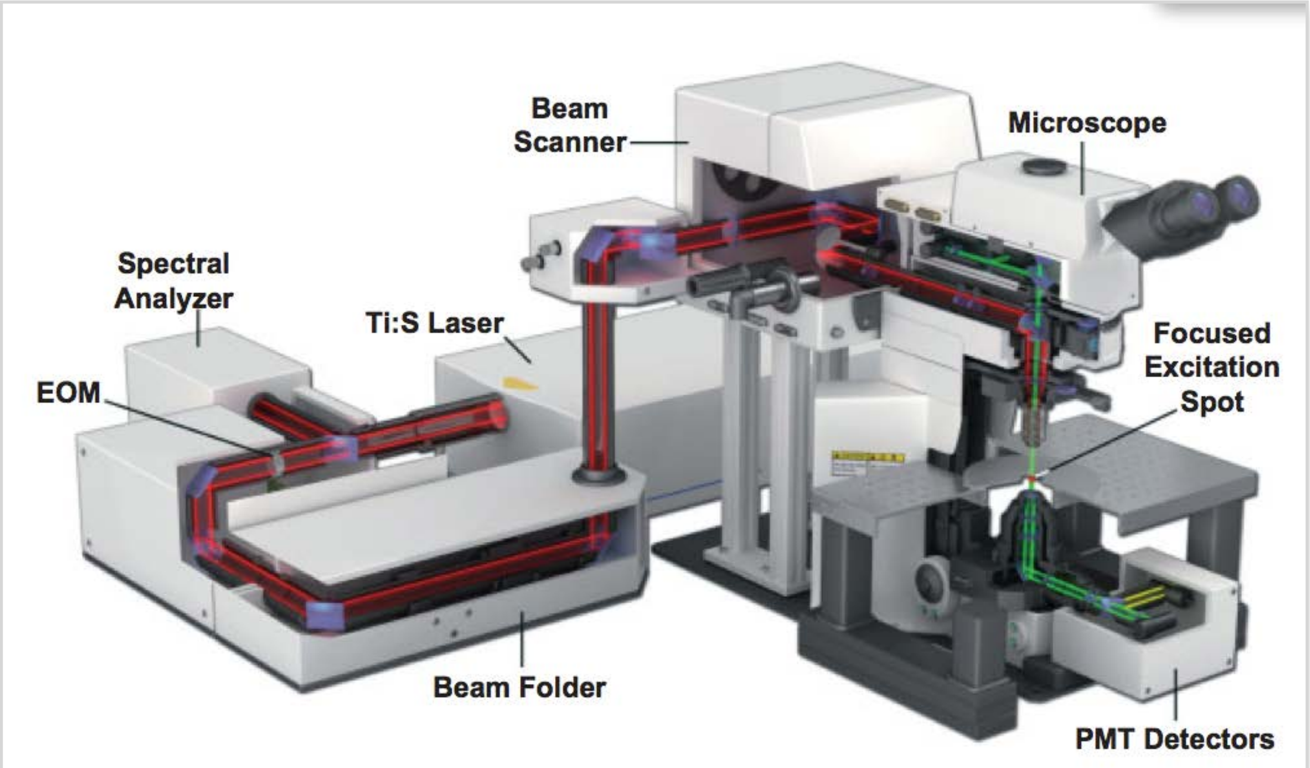


Two-Photon Microscopy:
pulsed near-infrared (NIR) laser & localised excitation of fluorescence

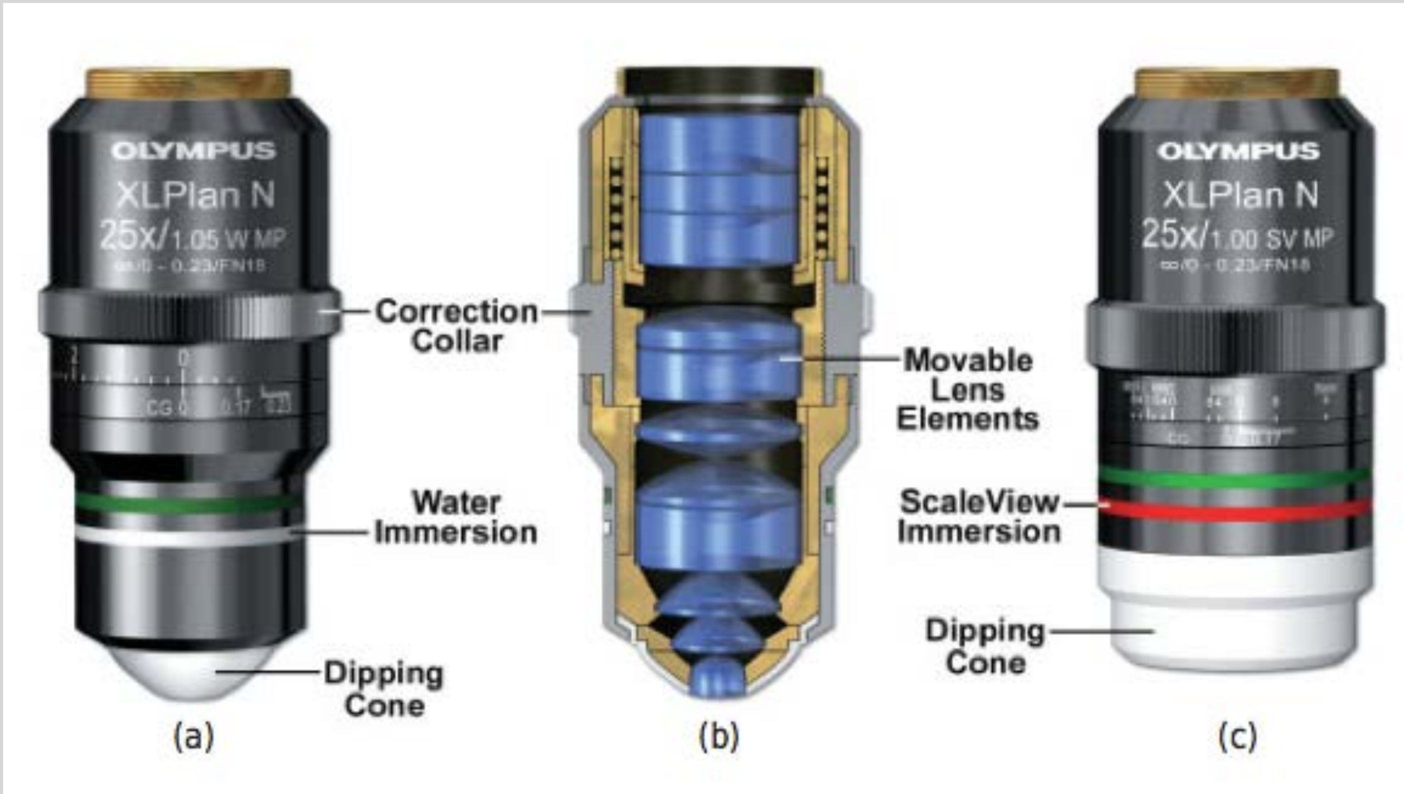
The mode-locked Ti:sapphire laser produces pulsed NIR light



Equipment configuration for a two-photon microscope



Objectives used for two-photon imaging



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Thank you!

Periklis (Laki) Pantazis